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- Abstracting causal Models

Example 3.2 [99 voters]

99 voters, each vote for/against

M_L: X_i, $i = 1 \to 99 \subseteq V$
$A_1, A_2, T \subseteq V$
U_i, $i = 1 \to 101 \subseteq U$

- $X_i \to$ Voter i's choice
  $1 \to$ for, $0 \to$ against
$A_i \to$ which advertisment is run
$T \to$ total # of votes for
- $U_i \to$ for $i = 1 \to 99$ determines how voter i votes as a function of which ads are run
U_i for $i = 100$ & $101 \to$ determine A_1 & A_2 respectively.

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* We have 3 clusters: G_1, G_2, G_3

$X_1 - X_{33}$, $X_{34} - X_{66}$, $X_{67} - X_{99}$

- Clusters represent sum of votes of each group
- replace $U_1 \cdots U_{99}$ by $U_1', U_2', \& U_3'$

T replaced with T' that just indicates who won.

- Only interventions allowed in M_L are to A_1 & A_2.

- we have a clear $\cancel{\ }^{I}$ mapping from *, & w is just the identity

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Example 3.15:

M_L: $\left.\begin{array}{l} V(\text{velocity}), H(\text{height}), \\ M(\text{mass}) \end{array}\right\} V$

$\left. U_V, U_H, U_M \right\} U$

F: $V = U_V$, $H = U_H$, $M = U_M$

M_H captures captures the objects current energy

M_H: $\underbrace{K(\text{Kinetic en.}) \,\&\, P(\text{potential en.})}_{V}$

$\left. U_K, U_P \right\} U$

$T: R(V_L) \to R(V_H)$ using:

$$T(v, h, m) = \left(\frac{1}{2}mv^2, 9.81mh\right)$$

$\omega_T(M \leftarrow m) = \emptyset$

$T(Rst(V_L, m)) = V_H$ by choosing v & h appropriately.

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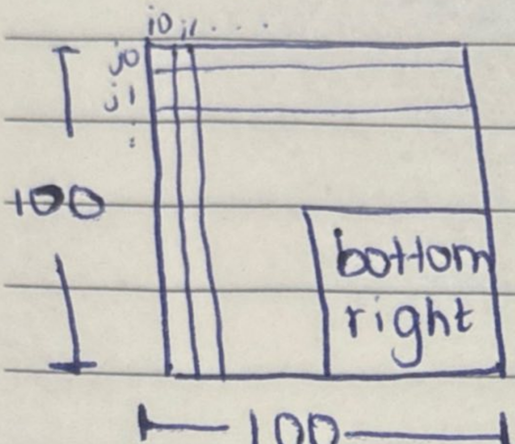
Example 3.16:



- each pixel can be black or white
- M_L: 10,000 end. variables X_{ij} & 10,000 exo. U_{ij}

- $X_{ij} = U_{ij}$
- we care about how many pixels in upper half are ~~grey~~ black & left half of the grid are also black.
- in M_H: U_H & $L_H \in V$ & $R(V) = \{0, \dots, 5000\}$
  - exogenous variable (m, m'), s.t $m \geq 0$ & $m' \leq 5{,}000$ & $|m - m'| \leq 2{,}500$

$W_T(\vec{X} \leftarrow \vec{x}) = 2$

* $UH \leftarrow m$ & $LH \leftarrow m'$, where m is the # of x_{ij} variables set to 1 with $1 \leq i \leq 50$ & m' is the number of x_{ij} variables set to 1 with $51 \leq j \leq 100$.

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